

Submission PHYS1-QUIZ-SUB01

Question PHYS1-QUIZ-Q01

scratch: show step

1. $v = v_i + at = 3 + 3(2) = 9 \text{ m/s}$

2. $\Delta x = v_i t + \frac{1}{2}at^2 = 3(2) + \frac{1}{2}(3)(2)^2 = 12 \text{ m}$

=> FINAL: $v = 9 \text{ m/s}$; displacement = 12 m (check units/interpretation)

Question PHYS1-QUIZ-Q02

scratch: show step

1. $N = mg = 68.6 \text{ N}$; friction = $\mu N = 13.7 \text{ N}$

2. $\Sigma F = 27 - 13.7 = 13.3 \text{ N}$

3. $a = \Sigma F / m = 1.90 \text{ m/s}^2$

=> FINAL: $a = 1.90 \text{ m/s}^2$; friction opposes motion (check units/interpretation)

Question PHYS1-QUIZ-Q03

scratch: show step

1. $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2 \cdot 9.8 \cdot 8}$

2. $v = 12.52 \text{ m/s}$; mass cancels

=> FINAL: $v = 12.52 \text{ m/s}$ (check units/interpretation)

Question PHYS1-QUIZ-Q04

scratch: show step

1. $p_{\text{total, before}} = p_{\text{total, after}}$

2. It applies when external impulse on the system is negligible.

=> FINAL: total momentum before equals after when external impulse is negligible (check units/interpretation)

Question PHYS1-QUIZ-Q05

scratch: show step

1. Impulse is area under $F(t)$: $J = F\Delta t = 6(0.5) = 3 \text{ N}\cdot\text{s}$.

2. $\Delta p = J$, so momentum changes by $3 \text{ kg}\cdot\text{m/s}$ in the force direction.

=> FINAL: impulse = $3 \text{ N}\cdot\text{s}$; momentum changes by $3 \text{ kg}\cdot\text{m/s}$ (check units/interpretation)